Code of the student

☐0	☐0	☐0	☐0	☐0	☐0	☐0	☐0
☐1	☐1	☐1	☐1	☐1	☐1	☐1	☐1
☐2	☐2	☐2	☐2	☐2	☐2	☐2	☐2
☐3	☐3	☐3	☐3	☐3	☐3	☐3	☐3
☐4	☐4	☐4	☐4	☐4	☐4	☐4	☐4
☐5	☐5	☐5	☐5	☐5	☐5	☐5	☐5
☐6	☐6	☐6	☐6	☐6	☐6	☐6	☐6
☐7	☐7	☐7	☐7	☐7	☐7	☐7	☐7
☐8	☐8	☐8	☐8	☐8	☐8	☐8	☐8
☐9	☐9	☐9	☐9	☐9	☐9	☐9	☐9

Surname, Name:

.....

DIAGNOSIS QUIZ MULTICAST

IMPORTANT REMARKS:

- It is important to fill the ID number of the student and the code of the test.
- IT IS COMPLETELY FORBIDDEN TO USE ANY KIND OF COMMUNICATIONS DEVICE.

Pregunta 1 According to RFC 2236, which of the following statements is FALSE regarding IGMPv2?

- ☐ IGMP is required to be implemented by all hosts wishing to receive IP multicasts
- ☐ IGMP messages are encapsulated in IP datagrams
- ☐ IGMP is a integral part of IP
- ☐ In the CORE, IGMP messages are sent between routers to construct a source-based tree from a Rendezvous Point to the source of information

Pregunta 2 To make a video transmission between **server** and **R2** you execute the following commands:

```
server# udp-sender --file=./big_664.mpg
--min-clients 1 --portbase 22345 --nopointopoint
--interface eth1 --ttl 1 --mcast-addr
232.43.211.234 --mcast-all-addr 225.1.2.3
R2# udp-receiver --file=big_664.mpg
--mcast-all-addr 225.1.2.3 --ttl 1 --interface
eth1 --portbase 22345
```

You have wireshark listening to all network interfaces. It is FALSE that:

- ☐ **server** sends IGMPv3 Membership Report messages to join to data group 232.43.211.234
- ☐ **R2** sends IGMPv3 Membership Report messages to join the data group 232.43.211.234
- ☐ **server** sends IGMPv3 Membership Report messages to join the control group 225.1.2.3
- ☐ **R2** sends IGMPv3 Membership Report messages to join the control group 225.1.2.3

Pregunta 3 You want to configure a BIDIRECTIONAL multicast static routing tree with multiple branches, to communicate **SimNet3** to both **SimNet0** and **SimNet5**. **SimNet3** is connected to **SimNet0** via a gre tunnel called `tunnel0`, and to **SimNet5** using `tunnel1`. Which of the following commands is NOT part of the configuration:

- ☐ R3# smcroute -a tunnel1 0.0.0.0 232.43.211.234 eth1
- ☐ R3# smcroute -a eth1 0.0.0.0 232.43.211.234 tunnel1
- ☐ R1# smcroute -a tunnel0 0.0.0.0 232.43.211.234 eth1
- ☐ R2# smcroute -a eth1 0.0.0.0 232.43.211.234 eth2

Pregunta 4 You have configured the multicast static routing tree to communicate **SimNet3** to **SimNet0** in a unidirectional way. These two islands are connected via a gre tunnel called `tunnel0`. You have wireshark listening to all network interfaces. You execute the following commands:

```
server#mcsender -t3 232.43.211.234:12345
host2#emcast -t3 232.43.211.234:12345
```

Then, it is FALSE that:

- ☐ In **host2** you will see the message `this is the test message from mclab/mcsender`
- ☐ Multicast packets do not reach destination because the TTL is too low
- ☐ All UDP packets sent are multicast packets from **server** and destination the multicast group 232.43.211.234
- ☐ This application is testing the multicast routing between **server** and **host2**

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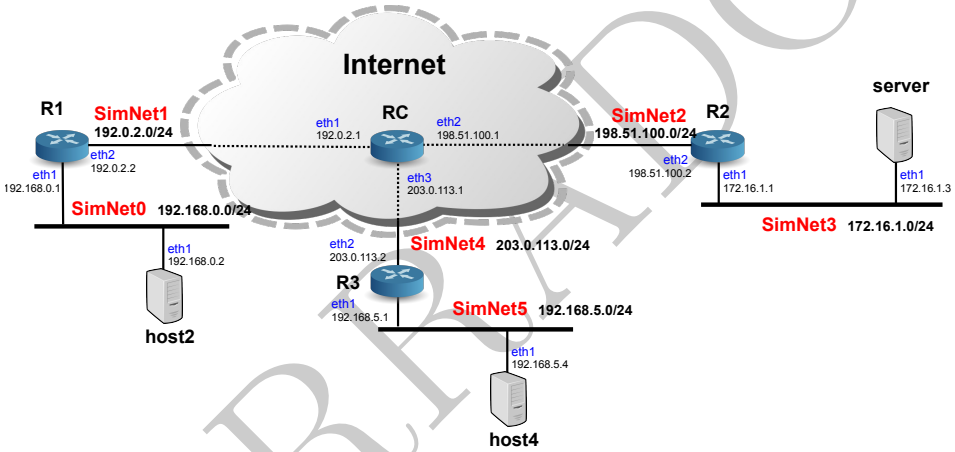


Fig. 1: Scenario Multicast

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